

BANGKOK METROPOLIS, Thailand

WMO: 484550

Lat: 13.7264N

Lon: 100.5600E

Elev: 14

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	68.7	71.4	53.9	61.9	75.6	57.0	69.4	76.8	9.9	84.5	9.0	84.3	1.4	60	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 12.7	(c) 97.2	(d) 80.4	(e) 95.9	(f) 80.2	(g) 94.7	(h) 79.8	(i) 83.2	(j) 91.6	(k) 82.5	(l) 90.8	(m) 81.9	(n) 90.0	(o) 4.5	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.1	161.8	87.3	80.5	158.3	86.5	79.9	155.3	86.1	47.2	91.9	46.3	91.0	45.7	90.3	92.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.3	(b) 9.0	(c) 7.6	(e) 64.5	(f) 100.4	(g) 3.6	(h) 1.7	(i) 61.9	(j) 101.6	(k) 59.8	(l) 102.6	(m) 57.8	(n) 103.5	(o) 55.1	(p) 104.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.9	82.1	84.4	86.6	88.5	87.2	86.1	85.2	85.0	84.3	84.2	83.8	81.7		
	DBStd	3.23	3.16	2.74	2.77	2.67	2.82	2.06	1.97	1.84	1.94	2.15	2.78	3.77		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	12742	994	964	1133	1154	1154	1082	1091	1084	1029	1060	1014	983		
	CDD65	7267	529	544	668	704	689	632	626	619	579	595	564	518		
	CDH74	89820	5755	6471	8627	9689	9273	8255	7844	7634	6817	7045	6715	5695		
(13)	CDH80	41497	2267	2840	4340	5449	4898	4026	3511	3326	2782	2903	2864	2291		
(14)	Wind	WSAvg	2.2	1.8	2.6	3.0	2.8	2.4	2.5	2.4	2.0	1.5	1.8	1.7		
(15) Precipitation	PrecAvg	60.8	0.6	0.9	1.5	3.1	8.5	6.3	6.5	8.3	13.5	10.5	2.1	0.5		
	PrecMax	88.2	5.2	4.9	6.9	14.2	21.8	16.2	13.7	22.6	26.6	22.1	11.6	4.1		
	PrecMin	34.5	0.0	0.0	0.0	0.0	1.2	1.1	1.3	2.0	6.5	0.9	0.0	0.0		
	PrecStd	12.0	1.1	1.2	1.7	3.0	5.0	3.4	2.9	4.2	4.3	4.4	2.2	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.6	95.0	97.2	99.3	99.1	96.4	95.0	94.7	94.4	94.1	95.0	94.2		
		MCWB	75.7	76.3	77.9	80.0	81.3	80.7	80.1	80.2	80.1	78.6	77.8	77.2		
	2%	DB	91.9	93.3	95.3	97.4	96.9	94.4	93.1	92.8	92.5	92.6	93.1	92.3		
		MCWB	75.7	77.2	78.8	80.6	81.3	80.3	80.0	79.9	79.8	78.6	77.6	76.5		
	5%	DB	90.7	92.0	94.0	96.1	95.2	92.8	91.7	91.3	91.1	91.4	91.6	90.8		
		MCWB	75.5	77.4	79.1	80.8	80.9	80.2	79.6	79.5	79.5	78.6	77.3	75.9		
	10%	DB	89.3	90.8	92.7	94.8	93.4	91.4	90.3	89.9	89.7	90.0	90.2	89.2		
		MCWB	75.1	77.3	79.1	80.6	80.5	79.9	79.2	79.1	79.2	78.6	77.0	75.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.7	81.8	82.5	84.5	84.2	82.8	82.7	83.2	81.9	81.7	81.6	80.3		
		MCDB	87.9	90.4	92.5	94.1	93.0	90.8	90.4	90.0	89.5	89.2	89.2	88.3		
	2%	WB	79.4	80.6	81.6	83.2	82.9	81.9	81.4	81.5	81.1	80.9	80.6	79.2		
		MCDB	86.3	88.8	90.6	92.2	91.2	89.7	89.0	88.5	88.4	88.0	87.8	87.1		
	5%	WB	78.7	79.8	80.9	82.5	82.3	81.5	80.7	80.7	80.6	80.3	79.9	78.4		
		MCDB	85.4	87.5	89.3	91.0	90.3	89.1	88.3	87.8	87.8	87.1	86.9	86.2		
	10%	WB	77.9	79.1	80.4	81.8	81.7	80.9	80.1	80.0	80.0	79.8	79.1	77.5		
		MCDB	84.6	86.3	88.3	90.2	89.8	88.4	87.4	87.0	86.8	86.4	86.1	85.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	13.4	12.5	12.7	12.0	11.2	10.5	10.6	11.2	11.5	12.4	13.9		
		MCDBR	15.4	14.2	13.7	13.7	13.5	12.9	12.2	12.3	12.9	13.1	13.6	14.4		
		MCWBR	4.9	4.2	3.8	3.9	4.1	4.3	4.3	4.3	4.3	3.9	4.1	4.5		
	5% WB	MCDBR	12.4	12.4	12.3	12.6	12.7	11.8	11.3	11.6	11.8	11.6	12.2	12.5		
		MCWBR	4.5	3.8	3.6	4.0	4.3	4.5	4.4	4.4	4.4	4.1	4.2	4.4		
(40) Clear-Sky Solar Irradiance	taub		0.545	0.612	0.638	0.576	0.499	0.466	0.458	0.460	0.484	0.580	0.526	0.511		
	taud		1.957	1.825	1.795	1.984	2.255	2.346	2.367	2.349	2.271	1.962	2.090	2.077		
	Ebn at Noon		239	230	228	241	256	262	265	268	263	235	244	246		
	Edh at Noon		57	68	71	59	44	40	39	41	44	58	50	50		
(44) All-Sky Solar Radiation	RadAvg		1617	1758	1805	1933	1763	1680	1605	1622	1568	1521	1565	1557		
(45)	RadStd		93	73	135	121	125	89	81	101	88	84	106	76		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.81	N/A	+0.59	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	+0.36	+0.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page